2

The Parametric Cure Model

2.1 Introduction

This chapter focuses on cure models that are fully parametrically specified. The parameters in such models can be readily estimated using the maximum likelihood method, and thus the properties of the estimates can be established using the maximum likelihood theory. Parametric cure models also allow extrapolation in prediction, which can be attractive in practice. The chapter is organized as follows. Section 2.2 introduces parametric mixture cure models, followed by some common estimation methods for the models in Section 2.3. Section 2.4 presents some non-mixture cure models as well as parametric transformation cure models that unify mixture cure models and other types of cure models. Model assessment issues such as choosing appropriate parametric distributions, selection between mixture versus non-mixture cure models, and goodness of fit of cure models are discussed in Section 2.5. Finally, some software packages that can fit parametric cure models and an example data set for cure models are presented in Section 2.6. This chapter focuses on independent and right-censored data only. The censoring is assumed to be independent and non-informative throughout this book unless otherwise specified.

2.2 Parametric Mixture Cure Models

As discussed in Chapter 1, we consider a sample drawn from a population that consists of two groups of subjects: cured subjects who respond well to their treatments and never experience the event of interest regardless of how long the follow-up is, and uncured subjects who will experience the event given a sufficient follow-up. It is natural to consider a mixture model for such a population. Let T be the time to the event of interest and Y be the cure indicator with $Y = 1$ if the subject is not cured and $Y = 0$ otherwise. We assume $P(T < \tau | Y = 1) = 1$, where τ is an upper bound of survival time of uncured subjects. Usually, τ is set to ∞ for theoretical convenience, and in this case, $Y = 0$ is not observable and a cured subject with $Y = 0$ is always censored. Thus for a subject with a right-censored time, the value of Y is not

observable and can be 1 or 0. However, it is possible to set τ to a finite value in practice, and in this case, $Y = 0$ is observable. For example, in a pregnancy study on time to abortion, $\tau = 20$ weeks is used because a pregnant woman is no longer considered susceptible to the event of abortion if the gestation age is 20 weeks or more.

Define $S_u(t)$ and $S_c(t)$ as the survival functions of the uncured and cured subpopulations respectively, i.e., $S_u(t) = P(T > t|Y = 1)$ and $S_c(t) = P(T > t|Y = 0)$ for any $t < \tau$. Following the definition of cured subjects, $S_c(t) = P(T > t|Y = 0) \equiv 1$ for any $t < \tau$, i.e., $S_c(t)$ is a degenerate survival function. The mixture cure model is defined by the following unconditional survival function of T for any $t < \tau$:

$$P(T > t) = S(t) = \pi S_u(t) + (1 - \pi)S_c(t) = \pi S_u(t) + 1 - \pi, \tag{2.1}$$

where $\pi = P(Y = 1)$ is the probability of being uncured. It is clear that the mixture cure model consists of two parts: one part is π, which is often referred to as the incidence part and describes the probability of being cured or uncured subjects, and the other part is $S_u(t)$, which is often referred to as the latency part and describes the distribution of the survival time of uncured subjects. This two-part structure facilitates the separate consideration of the effects of covariates on the cure probability and on the distribution of the survival time of uncured subjects, which renders appealing interpretation of covariate effects and easy generalization a mixture cure model to more complex situations. Because of the properties, the mixture cure model has been in use for near 70 years since it was first considered in the work of Boag (1949), and it is still attracting a great deal of attention.

The mixture cure model (2.1) is parametrically specified if both the latency submodel and incidence submodel are parametrically specified. In the following sections, we describe some parametric mixture cure models.

2.2.1 Parametric Incidence Submodel

Since Y is a binary variable, the only distribution for a binary variable is the Bernoulli distribution with the parameter π. Thus the specification of the incidence submodel of the mixture cure model is mainly focused on the specification of the functional form of the effect of $\boldsymbol{z}$, a vector of covariates (the first value of this vector is 1), and how it is linked to π. The most common parametric specification is the linear form of the effect $\boldsymbol{z}'\boldsymbol{\gamma}$ linked to π via a logit link:

$$\text{logit}[\pi(\boldsymbol{z})] = \boldsymbol{z}'\boldsymbol{\gamma}, \tag{2.2}$$

where $\text{logit}(\pi) = \log(\pi/(1 - \pi))$ and $\boldsymbol{\gamma}$ is a vector of the coefficients of the covariates in $\boldsymbol{z}$. For a particular covariate in $\boldsymbol{z}$, the corresponding coefficient in e^{γ} can be interpreted as the odds ratio of being uncured when the covariate is increased by 1 while other covariates are fixed.

Alternative link functions to link π and the linear predictor $\boldsymbol{z}'\boldsymbol{\gamma}$ can be considered. For example, one may consider the complementary log-log link function

$$\log[-\log(\pi(\boldsymbol{z}))] = -\boldsymbol{z}'\boldsymbol{\gamma}. \tag{2.3}$$

For a particular covariate in $\boldsymbol{z}$, the corresponding coefficient in e^{γ} can be interpreted as the relative log risk (not relative risk) of being uncured when the covariate is increased by 1 while other covariates are fixed. The probit link function $\Phi^{-1}[\pi(\boldsymbol{z})] = \boldsymbol{z}'\boldsymbol{\gamma}$, where $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution, is another link function that can be considered. However, due to unavailability of the closed form in $\Phi(\cdot)$, there is no straightforward interpretation of $\boldsymbol{\gamma}$ under this link function.

The logit and probit links are symmetric in the sense that modeling $\pi(\boldsymbol{z})$ and $1 - \pi(\boldsymbol{z})$ are equivalent and the coefficients in $\boldsymbol{\gamma}$ only differ by a sign. The complementary log-log link, on the other hand, is asymmetric. That is, it approaches 0 slowly and 1 fast, and its models for $\pi(\boldsymbol{z})$ and $1 - \pi(\boldsymbol{z})$ are not equivalent. The differences between the links functions can be seen in Figure 2.1, where $\boldsymbol{z} = (1, z)'$ and a) logit link with $\pi(\boldsymbol{z}) = e^z/(1 + e^z)$; b) complementary log-log link with $\pi(\boldsymbol{z}) = \exp(-e^{-0.3665-z})$; c) probit link with $\pi(\boldsymbol{z}) = \Phi(0.6z)$ is plotted against z. All the functions leads to $\pi(\boldsymbol{z}) = 0.5$ when $z = 0$ and the parameters in the complementary log-log and probit links are chosen to produce curves close to the curve from the logit link. The figure shows the symmetry in the logit and probit links and minor differences between the two curves with extreme values of z. The complementary log-log link, on the other hand, is clearly not symmetric and displays noticeable differences when $z < 0$ while the curve fits the logit curve for $z > 0$.

It can be challenging to choose an appropriate link for a given data set in practice. When the sample size is large, one may consider to use

$$\pi(\boldsymbol{z}) = \left(1 + \lambda e^{-\boldsymbol{\gamma}'\boldsymbol{z}}\right)^{-1/\lambda}, \quad 0 \le \lambda \le 1 \tag{2.4}$$

based on the Box-Cox transformation (Box and Cox, 1964). It is easy to see that $\pi(\boldsymbol{z}) = e^{\boldsymbol{\gamma}'\boldsymbol{z}}/(1+e^{\boldsymbol{\gamma}'\boldsymbol{z}})$ when $\lambda = 1$ and $\pi(\boldsymbol{z}) = \exp(-e^{-\boldsymbol{\gamma}'\boldsymbol{z}})$ when $\lambda \to 0$. Therefore (2.4) includes both the logit link and the complementary log-log link as special cases. An estimate value of λ and its standard error may produce useful information to determine which of the two link functions is adequate for the data if it is not clear which link function should be used.

2.2.2 Parametric Latency Submodel

A parametric specification of the latency submodel of the mixture cure model involves a parametric specification of the distribution of $T|Y = 1$ and a parametric specification of the effect of $\boldsymbol{x}$, a vector of covariates, on the distribution. This is often achieved by writing $S_u(t)$ as a function of a parametrically

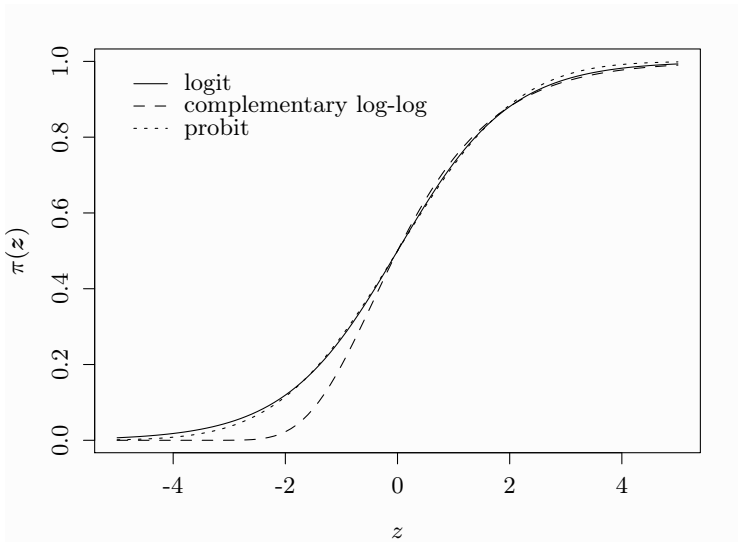

FIGURE 2.1
Comparison of three link functions for the incidence part of the mixture cure model.

specified baseline survival function $S_{u0}(t)$ and $\boldsymbol{\beta}'\boldsymbol{x}$, where $S_u(t) \equiv S_{u0}(t)$ when $\boldsymbol{x} = 0$, and $\boldsymbol{\beta}$ is a vector of the coefficients of the covariates in $\boldsymbol{x}$. The covariate vectors $\boldsymbol{x}$ and $\boldsymbol{z}$ may or may not include the same covariates.

2.2.2.1 Parametric PH Latency Submodel

One method to specify the latency part is to use the proportional hazards (PH) assumption:

$$S_u(t) = S_u(t|\boldsymbol{x}) = S_{u0}(t)^{\exp(\boldsymbol{x}'\boldsymbol{\beta})} \tag{2.5}$$

and $S_{u0}(t)$ is a survival function from a parametric distribution. For example, if $S_{u0}(t) = e^{-\lambda t}$, the survival function from an exponential distribution with rate λ, then $S_u(t|\boldsymbol{x}) = e^{-\lambda \exp(\boldsymbol{x}'\boldsymbol{\beta})t}$ is also a survival function from an exponential distribution with rate $\lambda \exp(\boldsymbol{x}'\boldsymbol{\beta})$. Similarly, if $S_{u0}(t) = e^{-\lambda t^p}$, the survival function from a Weibull distribution with shape parameter p and scale parameter λ, then $S_u(t|\boldsymbol{x}) = e^{-\lambda \exp(\boldsymbol{x}'\boldsymbol{\beta})t^p}$ is also the survival function from a Weibull distribution with shape parameter p and scale parameter $\lambda \exp(\boldsymbol{x}'\boldsymbol{\beta})$. Note that $\boldsymbol{\beta}$ in the above cases should not contain an intercept because it will not be identifiable from the scale parameter λ in the baseline distributions. The interpretation of $\boldsymbol{\beta}$ is similar to Cox's PH model. That is, for any covariate in $\boldsymbol{x}$, the corresponding coefficient in $\boldsymbol{\beta}$ is the log-hazard ratio if the covariate increases by 1 while other covariates are controlled.

The parametric latency submodel with the PH assumption is not widely used except for the cases discussed above because a semiparametric estimation method is available for the latency submodel under the PH assumption without specifying the baseline distribution parametrically, making a parametric PH model as the latency submodel less attractive.

2.2.2.2 Parametric AFT Latency Submodel

The parametric submodel for the latency part is more popular under the so-called accelerated failure time (AFT) assumption (Cox and Oakes, 1984), which can be written as follows:

$$S_u(t|\boldsymbol{x}) = S_{u0}(te^{-\boldsymbol{x}'\boldsymbol{\beta}}), \tag{2.6}$$

where $S_{u0}(t)$ is the survival function of a baseline distribution. If

$$\log(T|Y=1) = \boldsymbol{x}'\boldsymbol{\beta} + \sigma\epsilon \tag{2.7}$$

with σ as a scale parameter and ϵ as an error term satisfying $P(e^{\sigma\epsilon} > t) = S_{u0}(t)$, then $T|Y=1$ will follow the AFT model (2.6). Unlike the PH assumption, the AFT assumption allows direct interpretation of the effects of $\boldsymbol{x}$ on the $\log T|Y=1$ scale. Most of the parametric latency submodels based on the AFT assumption is via (2.7) by assuming a parametric distribution for $e^{\sigma\epsilon}$. For example, if e^{ϵ} follows the exponential distribution with mean 1 or if ϵ follows the extreme value distribution with survival function $P(\epsilon > s) = \exp(-e^s)$, then $S_u(t|\boldsymbol{x}) = \exp\left[-\left(te^{-\boldsymbol{x}'\boldsymbol{\beta}}\right)^{1/\sigma}\right]$. The corresponding mixture cure model is a Weibull AFT mixture cure model. It is well known such a latency submodel also satisfies the PH assumption and thus is also a Weibull PH mixture cure model (Farewell, 1982, 1986). If $\sigma = 1$, the model reduces to the exponential mixture cure model and is well studied in the early 1990s. See Maller and Zhou (1996) for the details.

There are many parametric distributions that can be considered for $e^{\sigma\epsilon}$ or ϵ when the Weibull baseline distribution is not appropriate. If the standard normal distribution is assumed for ϵ, then $S_u(t|\boldsymbol{x})$ is from the lognormal distribution (Gamel and McLean, 1994). Some multi-parameter distributions are proposed for ϵ as well, such as the extended generalized log-gamma distribution (Yamaguchi, 1992) with a density function

$$f_\epsilon(\epsilon) = \begin{cases} \frac{|q|}{\Gamma(q^{-2})}(q^{-2})^{q-2}\exp\left[q^{-2}(q\epsilon - e^{q\epsilon})\right] & \text{if } q \neq 0 \\ \frac{1}{\sqrt{2\pi}}\exp(-\epsilon^2/2) & \text{if } q = 0, \end{cases} \tag{2.8}$$

or a generalized F distribution with density function

$$f_\epsilon(\epsilon; s_1, s_2) = \frac{(s_1 e^{\epsilon}/s_2)^{s_1}}{(1 + s_1 e^{\epsilon}/s_2)^{s_1+s_2} B(s_1, s_2)}, \tag{2.9}$$

where $s_1 > 0$ and $s_2 > 0$ and $\Gamma(\cdot)$ and $B(\cdot,\cdot)$ are gamma and beta functions (Peng et al., 1998). The complexity of the multi-parameter distributions makes the interpretation of the parameters in the latency submodel difficult. However, the fact that they include many simpler distributions, such as the Weibull, lognormal, and log-logistic distributions, as special cases makes them suitable to test the adequacy of the simpler distributions as the distribution for the latency part. See Section 2.5 for more details.

2.2.2.3 Other Parametric Latency Submodels

In addition to the PH and AFT assumptions above, there are other less known assumptions that can be considered for the latency submodel. The accelerated hazards (AH) assumption was first considered by Chen and Wang (2000) for survival data. It is in the following form when used as the latency submodel:

$$S_u(t|\boldsymbol{x}) = S_{u0}(te^{-\boldsymbol{x}'\boldsymbol{\beta}})^{\exp(\boldsymbol{x}'\boldsymbol{\beta})}, \tag{2.10}$$

where $S_{u0}(t)$ is a parametric baseline survival function. It may be easier to see how $\boldsymbol{x}$ affects the latency distribution under the AH assumption by the corresponding hazard function: $h_u(t|\boldsymbol{x}) = h_{u0}(te^{-\boldsymbol{x}'\boldsymbol{\beta}})$. Thus for a non-zero coefficient in $\boldsymbol{\beta}$, the corresponding covariate accelerates/decelerates the baseline hazard function. Due to the nature of the AH assumption, the baseline survival function can be from any distribution that is suitable for modeling time except for the exponential distribution. Unlike the PH and AFT assumptions above, the AH assumption allows a gradual effect of $\boldsymbol{x}$ on the distribution of T for uncured subjects. That is, if the baseline hazard function is monotone but not a constant, the hazard functions of two groups differ only when $t > 0$, and the larger the time t, the greater the differences in hazard.

The proportional odds (PO) assumption can also be considered in the latency submodel as follows:

$$S_u(t|\boldsymbol{x}) = \frac{1}{1 + [S_{u0}(t)^{-1} - 1]e^{-\boldsymbol{\beta}'\boldsymbol{x}}} \tag{2.11}$$

where $S_{u0}(t)^{-1} - 1$ is a parametric baseline odds function. Contrary to the AH assumption, the PO assumption implies that the hazard ratio of the uncured subjects approaches one as $t \to \infty$. That is, the differences in hazard will fade away under the PO assumption, making it suitable to model the effect of a covariate that diminishes as the survival time increases.

The four assumptions above are the examples of ways to specify the effects of $\boldsymbol{x}$ on $S_u(t)$ in the latency part. Determining an appropriate assumption among them for a given data set is important but not trivial. Hutton and Monaghan (2002) discussed issues of misspecification between PH and AFT assumptions for models without a cure fraction. Figure 2.2 displays two hazard functions from two groups under the four assumptions when one hazard function is a linear function of time. One may choose an assumption based on how the hazard ratio changes. For example, if the hazard ratio is a constant, the

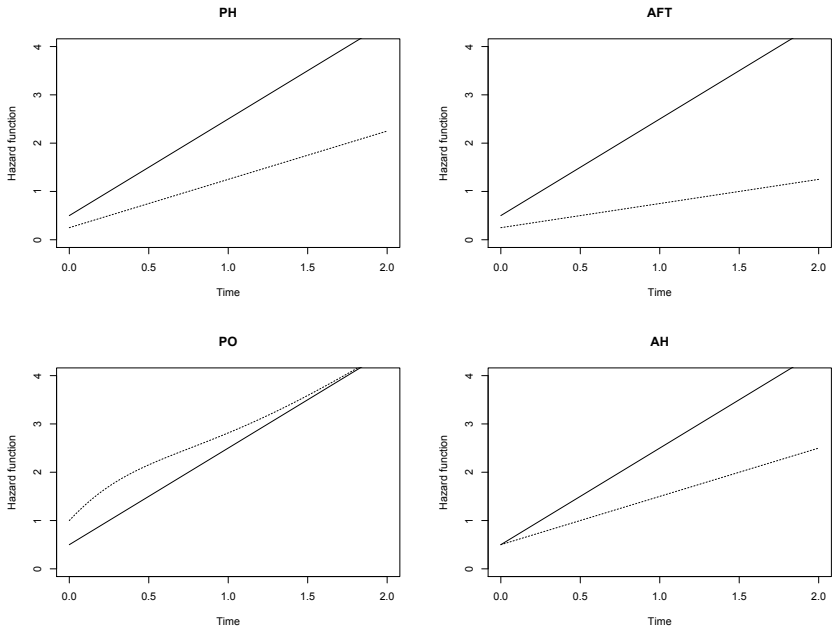

FIGURE 2.2
Examples of hazard functions of two groups under the four assumptions.

PH assumption would be appropriate. If the hazard difference between groups diminishes over time, the PO assumption may be a reasonable choice. Or one can determine a model based on whether and how the hazard functions cross. See discussions in Zhang and Peng (2009b) for some details. Unfortunately, the information about the hazard ratio for a covariate is often not available. The missing values in Y makes it infeasible to use any methods that were proposed to differentiate these models for data without cured subjects.

2.3 Model Estimation

2.3.1 Direct Maximization of Observed Likelihood Function

Suppose that the observed data of T and its censoring indicator δ, covariate $\boldsymbol{x}$, and $\boldsymbol{z}$ for n subjects are denoted as $(t_i, \delta_i, \boldsymbol{x}_i, \boldsymbol{z}_i)$, $i = 1, \ldots, n$, where $\delta_i = 1$ if t_i is uncensored and $\delta_i = 0$ if t_i is censored. Let $\boldsymbol{\alpha}$ be a vector of unknown parameters in the parametric baseline survival function $S_{u0}(\cdot)$, and

$\boldsymbol{\theta} = (\boldsymbol{\gamma}, \boldsymbol{\beta}, \boldsymbol{\alpha})'$. The log-likelihood function of the mixture cure model is

$$\ell(\boldsymbol{\theta}) = \log \prod_{i=1}^{n} [\pi(\boldsymbol{z}_i) f_u(t_i|\boldsymbol{x}_i)]^{\delta_i} [1 - \pi(\boldsymbol{z}_i) + \pi(\boldsymbol{z}_i) S_u(t_i|\boldsymbol{x}_i)]^{1-\delta_i}. \tag{2.12}$$

The log-likelihood function can be maximized directly using the Newton-Raphson method. Let

$$\boldsymbol{U}(\boldsymbol{\theta}) = \frac{\partial \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \begin{pmatrix} \frac{\partial \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\gamma}} \\ \frac{\partial \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\beta}} \\ \frac{\partial \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\alpha}} \end{pmatrix}, I(\boldsymbol{\theta}) = -\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} = -\begin{pmatrix} \frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\gamma} \partial \boldsymbol{\gamma}'} & \frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\gamma} \partial \boldsymbol{\beta}'} & \frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\gamma} \partial \boldsymbol{\alpha}'} \\ \frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\gamma}'} & \frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} & \frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\alpha}'} \\ \frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\alpha} \partial \boldsymbol{\gamma}'} & \frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\alpha} \partial \boldsymbol{\beta}'} & \frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\alpha} \partial \boldsymbol{\alpha}'} \end{pmatrix}.$$

Then the maximum likelihood estimate $\widehat{\boldsymbol{\theta}}$ of $\boldsymbol{\theta}$ can be obtained as a result of converged iteration steps

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} + I^{-1}(\boldsymbol{\theta}^{(k)}) \boldsymbol{U}(\boldsymbol{\theta}^{(k)}),$$

and its variance can be approximated by $I^{-1}(\widehat{\boldsymbol{\theta}})$. This approach relies on the availability of the first and second derivatives of the log-likelihood function with respect to all the parameters in the model. This can be an issue when dealing with some multi-parameter distributions such as the generalized gamma or F distribution. Peng et al. (1998) suggested to maximize the log-likelihood function directly using the simulated annealing algorithm based on the downhill simplex method (Press et al., 1992), which does not depend on any derivatives, or use the algorithm for $\boldsymbol{\alpha}$ and the Newton-Raphson method for $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$.

2.3.2 Estimation via EM Algorithm

The EM algorithm (Dempster et al., 1977) is another method that can be conveniently used to obtain the maximum likelihood estimates of the parameters in the mixture cure models. Let y_i be the value of Y for subject i. Then $y_i = 1$ if $\delta_i = 1$, and is usually latent or unknown with $\delta_i = 0$. The unknown status of y_i is due to the fact that a subject will be censored if either the subject is cured or the subject is not cured but the subject's failure time T is greater than the censoring time C. There is no information to tell whether the subject is cured or not given the censored failure time.

If all values of y_i's are available, the corresponding complete log-likelihood function based on the complete data $(t_i, \delta_i, \boldsymbol{x}_i, \boldsymbol{z}_i, y_i)$, $i = 1, \ldots, n$ is (up to an additive constant)

$$\begin{aligned} \ell^c(\boldsymbol{\gamma}, \boldsymbol{\beta}, \boldsymbol{\alpha}) &= \log \prod_{i=1}^{n} [\pi(\boldsymbol{z}_i) f_u(t_i|\boldsymbol{x}_i)^{\delta_i} S_u(t_i|\boldsymbol{x}_i)^{1-\delta_i}]^{y_i} [1 - \pi(\boldsymbol{z}_i)]^{1-y_i} \\ &= \ell_1(\boldsymbol{\gamma}) + \ell_2(\boldsymbol{\beta}, \boldsymbol{\alpha}), \end{aligned}$$

where

$$\ell_1(\boldsymbol{\gamma}) = \sum_{i=1}^{n} \{y_i \log[\pi(\boldsymbol{z}_i)] + (1 - y_i) \log[1 - \pi(\boldsymbol{z}_i)]\},$$

$$\ell_2(\boldsymbol{\beta}, \boldsymbol{\alpha}) = \sum_{i=1}^{n} y_i \{\delta_i \log[f_u(t_i|\boldsymbol{x}_i)] + (1 - \delta_i) \log[S_u(t_i|\boldsymbol{x}_i)]\}.$$

Let

$$w_{0i}(t) = \frac{\pi(\boldsymbol{z}_i) S_u(t|\boldsymbol{x}_i)}{1 - \pi(\boldsymbol{z}_i) + \pi(\boldsymbol{z}_i) S_u(t|\boldsymbol{x}_i)} \tag{2.13}$$

be the probability of a subject with covariates $\boldsymbol{x}_i$ and $\boldsymbol{z}_i$ to be uncured conditional on surviving at t under the mixture cure model (2.1). Given the estimates of $\boldsymbol{\gamma}, \boldsymbol{\beta}, \boldsymbol{\alpha}$ in the $(k-1)$th iteration of the EM algorithm, denoted as $\boldsymbol{\gamma}^{(k-1)}, \boldsymbol{\beta}^{(k-1)}, \boldsymbol{\alpha}^{(k-1)}$, the E-step in the kth iteration of the EM algorithm calculates the posterior expectation of y_i as

$$w_i^{(k)} = \delta_i + (1 - \delta_i) w_{0i}(t_i) \big|_{\boldsymbol{\gamma}=\boldsymbol{\gamma}^{(k-1)}, \boldsymbol{\beta}=\boldsymbol{\beta}^{(k-1)}, \boldsymbol{\alpha}=\boldsymbol{\alpha}^{(k-1)}} \tag{2.14}$$

following the property of y_i above and Bayes' theorem. It can be interpreted as the probability the subject being uncured given the observed data for that subject and current parameter estimates.

The M-step in the kth iteration of the EM algorithm maximizes ℓ_1 and ℓ_2 after replacing y_i in ℓ_1 and ℓ_2 with $w_i^{(k)}$. That is, the M-step maximizes

$$\ell_1(\boldsymbol{\gamma}) = \sum_{i=1}^{n} \left\{ w_i^{(k)} \log[\pi(\boldsymbol{z}_i)] + (1 - w_i^{(k)}) \log[1 - \pi(\boldsymbol{z}_i)] \right\}, \tag{2.15}$$

$$\ell_2(\boldsymbol{\beta}, \boldsymbol{\alpha}) = \sum_{i=1, w_i^{(k)}>0}^{n} w_i^{(k)} \{\delta_i \log[f_u(t_i|\boldsymbol{x}_i)] + (1 - \delta_i) \log[S_u(t_i|\boldsymbol{x}_i)]\} \tag{2.16}$$

to update the estimates of $\boldsymbol{\gamma}, \boldsymbol{\beta}, \boldsymbol{\alpha}$. The updated estimates of $\boldsymbol{\gamma}, \boldsymbol{\beta}, \boldsymbol{\alpha}$, denoted as $\boldsymbol{\gamma}^{(k)}, \boldsymbol{\beta}^{(k)}, \boldsymbol{\alpha}^{(k)}$, will be compared to $\boldsymbol{\gamma}^{(k-1)}, \boldsymbol{\beta}^{(k-1)}, \boldsymbol{\alpha}^{(k-1)}$ to determine the convergence of the algorithm. If not, the E-step and M-step iterate until a convergence is achieved.

It is easy to see that (2.15) can be viewed as the log-likelihood function from a logistic regression with $(w_1^{(k)}, \ldots, w_n^{(k)})$ as values of the response variable. Thus, it can be maximized by the Newton-Raphson method or using existing software for the logistic regression (Peng, 2003b). For (2.16), it is similar to the log-likelihood function for censored data except the multiplicative term $w_i^{(k)}$. Therefore, it can be treated as a weighted log-likelihood function for censored data without a cure fraction and can be maximized by the Newton-Raphson method or by existing software for classic survival models that accept case weights. For example, one can assume that

$S_{u0}(t) = \exp(-\lambda t)$ is the survival function of the exponential distribution with rate λ in (2.5). The log-likelihood function reduces to $\ell_2(\boldsymbol{\beta}, S_{u0}) = \ell_2(\boldsymbol{\beta}, \lambda) = \sum_{i=1, w_i^{(k)}>0}^{n} \left\{ \delta_i \log \lambda + \delta_i \boldsymbol{\beta}' \boldsymbol{x}_i - \lambda t_i e^{\log w_i^{(k)} + \boldsymbol{\beta}' \boldsymbol{x}_i} \right\}$. It can be maximized by either the Newton-Raphson method or an existing software package to fit the exponential distribution to data that allows case weight.

The M-step can also be completed using the Monte Carlo EM algorithm of Tanner and Wong (1987) by simulating y_i's from w_i's in (2.14). See Lam et al. (2005) and Xu and Zhang (2010) for details.

These features demonstrate that the EM algorithm for the mixture cure model is not only an alternative method to obtain the maximum likelihood estimates of the parameters in the model, but also a bridge to connect the mixture cure models with existing classical survival models, which makes it possible to use existing methods for the classical survival models to estimate the parameters in the mixture cure models.

One challenge in using EM algorithm is that the variances of the estimates are not directly available at the convergence of the EM algorithm. For the parametric mixture cure model, the methods proposed by Louis (1982) and Oakes (1999) may be considered. These methods depend on availability of derivatives of ℓ_1 and ℓ_2 with respect to the parameters in the model, which can be challenging for some parameters in the specified distribution in ℓ_2. An alternative approach is to use the bootstrap method (Davison and Hinkley, 1997; Li and Datta, 2001) to estimate the standard errors of the estimates.

2.4 Non-Mixture Cure Models

2.4.1 Proportional Hazards Cure Model

Another approach to define a cure model is to based on a theory of tumor kinetics in cancer studies. The theory assumes (Yakovlev et al., 1996; Tsodikov, 2001) that each subject has N cancer cells at the baseline. A cured subject has $N = 0$ cancer cells while an uncured subject has $N > 0$ cancer cells, and the uncured subject will develop cancer if one of the $N > 0$ cancer cells develops a detectable cancer mass. That is,

$$T = \min\{\tilde{T}_1, \ldots, \tilde{T}_N\} = \tilde{T}_{(1)}, \tag{2.17}$$

where $\tilde{T}_i$'s are i.i.d. latent event times or activation times for cancer cells to develop a detectable cancer mass, and $\tilde{T}_{(1)}$ is the first order statistic of $\tilde{T}_1, \ldots, \tilde{T}_N$. This is also called the first-activation scheme and is one of a number of schemes developed in Cooner et al. (2007) that may be suitable for tumor kinetics. The schemes involve different distributions for N and $\tilde{T}_i$ and different order statistics $\tilde{T}_{(r)}$, $1 \le r \le N$, to define the failure time T, and they lead to

different cure models. The most popular model is based on the first-activation scheme with N following a Poisson distribution with mean $\exp(\gamma_0+\boldsymbol{z}'\boldsymbol{\gamma})$ that is independent of $\tilde{T}_i$'s. If $\tilde{T}_i \sim F^H(t)$, a proper cumulative distribution function, then unconditional survival function of T is

$$\begin{aligned} P(T>t) = S(t|\boldsymbol{z}) =& P(N=0) + P(\tilde{T}_1 > t, \ldots, \ldots, \tilde{T}_N > t, N \geq 1) \\ =& \exp(-e^{\gamma_0+\boldsymbol{z}'\boldsymbol{\gamma}}) + \sum_{k=1}^{\infty} \frac{[S^H(t)e^{\gamma_0+\boldsymbol{z}'\boldsymbol{\gamma}}]^k}{k!} \exp(-e^{\gamma_0+\boldsymbol{z}'\boldsymbol{\gamma}}) \\ =& \exp[-e^{\gamma_0+\boldsymbol{z}'\boldsymbol{\gamma}} F^H(t)], \end{aligned} \tag{2.18}$$

where $S^H(t) = 1 - F^H(t)$. The model can be rewritten as $S(t|z) = [e^{-\exp(\gamma_0)F^H(t)}]^{\exp(\boldsymbol{z}'\boldsymbol{\gamma})}$ or $h(t|\boldsymbol{z}) = [e^{\gamma_0} f^H(t)]\exp(\boldsymbol{z}'\boldsymbol{\gamma})$, where $f^H(t) = dF^H(t)/dt$. It is obviously similar to the classic PH model except that the baseline survival function counterpart of the PH model in model (2.18) is an improper survival function satisfying $\lim_{t\to\infty} \exp[-e^{\gamma_0}F^H(t)] = \exp(-e^{\gamma_0}) \in (0, 1)$. Thus, we refer to this model as the proportional hazards cure (PHC) model. Since the baseline cumulative hazard counterpart of the PH model in model (2.18), $e^{\gamma_0}F^H(t)$, is bounded, model (2.18) is also called the bounded cumulative hazard cure model.

As a comparison, the mixture cure model does not have a PH structure when the latency does not depend on any covariates. This can be seen from the fact that

$$h(t|\boldsymbol{z}) = \frac{f(t|\boldsymbol{z})}{S(t|\boldsymbol{z})} = \frac{\pi(\boldsymbol{z})f_u(t)}{\pi(\boldsymbol{z})S_u(t) + 1 - \pi(\boldsymbol{z})} = w_0(t, \boldsymbol{z})h_u(t).$$

As in the discussion for (2.13), $w_0(t, \boldsymbol{z})$ is a conditional probability of being uncured at t and is generally not independent of t. Thus, the model does not have the PH structure in general (the mixture cure model may behave like the PH model with $h(t|\boldsymbol{z}) = \pi(\boldsymbol{z})h_u(t)$ when $t \to 0$ or $h(t|\boldsymbol{z}) = \frac{\pi(\boldsymbol{z})}{1-\pi(\boldsymbol{z})} f_u(t)$ when $t \to \infty$).

If there is no covariate present and $F^H(t)$ is the distribution function of the exponential distribution, then $S(t|\boldsymbol{z})$ is a survival function from an improper generalized Gompertz distribution. This particular parametric PHC model was discussed in detail in the literature (Cantor and Shuster, 1992; Yakovlev et al., 1994b; Cantor, 2001).

The PHC model can be written in a form of a mixture cure model (2.1) where $\pi = 1 - \exp(-e^{\gamma_0+\boldsymbol{z}'\boldsymbol{\gamma}})$ and

$$S_u(t) = \frac{\exp[e^{\gamma_0+\boldsymbol{z}'\boldsymbol{\gamma}}F^H(t)] - 1}{\exp[e^{\gamma_0+\boldsymbol{z}'\boldsymbol{\gamma}}] - 1}.$$

The incidence part is essentially the complementary log-log model (2.3). However, the latency part does not have a simple interpretation of covariate effects as in the mixture cure models discussed in Section 2.2. Moreover, it is clear

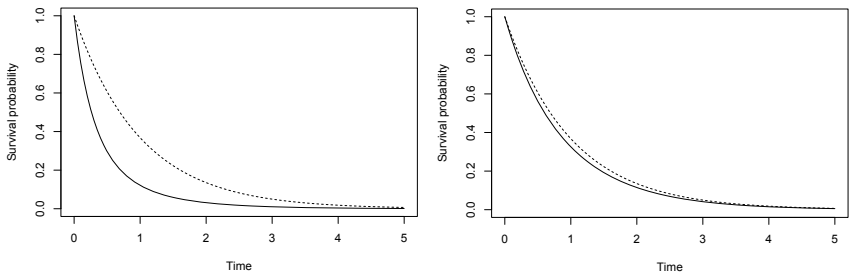

FIGURE 2.3
$S_u(t)$ (solid line) and $S^H(t)$ (dotted line) when the cure rate is 7% (left) and 69% (right).

that for any $\boldsymbol{z}$, we have $S_u(t) \leq S^H(t)$ and equality occurs when $e^{\gamma_0+\boldsymbol{z}'\boldsymbol{\gamma}} \to 0$. It implies that $S^H(t)$ cannot be viewed as a survival function for uncured subjects, unless the cure rate is close to 1. Figure 2.3 demonstrates the difference between $S_u(t)$ and $S^H(t)$ when the cure rate increases (Peng and Xu, 2012).

As an alternative to the mixture cure model, the PHC model received a great deal of interest in the last decade due to its similarity to the classical PH model. However, the proportionality restriction in $\boldsymbol{z}$ in the model also limits the wide applications of the model in practice. The restriction can be relaxed by including covariate effects in $F^H(t)$, such as under the proportional hazards assumption

$$1 - F^H(t|\boldsymbol{x}) = S^H(t|\boldsymbol{x}) = S_0^H(t)^{\exp(\boldsymbol{x}'\boldsymbol{\beta})}, \tag{2.19}$$

where $S_0^H(t)$ is a baseline survival function (Tsodikov, 2002). This is the so-called PHPH cure model. The inclusion of $\boldsymbol{x}$ in $F^H(t)$ widens the applicability of the PHC model, because there are now two sets of parameters to describe the effects of the covariates. But the covariate effects in $S_u(t)$ are difficult to interpret compared to the PH mixture cure model because both effects of $\boldsymbol{z}$ and $\boldsymbol{x}$ appear in the latency part.

Let $f(t|\boldsymbol{z}) = e^{\gamma_0+\boldsymbol{z}'\boldsymbol{\gamma}} f^H(t) \exp[-e^{\gamma_0+\boldsymbol{z}'\boldsymbol{\gamma}} F^H(t)]$ be the unconditional density function corresponding to (2.18) and $F^H(t)$ is parametrically specified up to a few unknown parameters $\boldsymbol{\alpha}$. The log-likelihood function for the model is

$$\ell(\gamma_0, \boldsymbol{\gamma}, \boldsymbol{\alpha}) = \log \prod_{i=1}^{n} [f(t_i|\boldsymbol{z}_i)]^{\delta_i} [S(t_i|\boldsymbol{z}_i)]^{1-\delta_i}.$$

The standard Newton-Raphson method discussed in Section 2.3.1 can be used to maximize the likelihood function to obtain the maximum likelihood estimates of the model parameters. See Asselain et al. (1996) and Tsodikov et al. (1998) for some details of the model estimation and applications.

2.4.2 Cure Models Based on Tumor Activation Scheme

The PHC model can be viewed as a result of the latent first-activation scheme described as (2.17), which is one of a few latent activation schemes proposed. Other latent activation schemes are discussed in detail in Section 7.2. As indicated by Tucker and Taylor (1996), however, the interpretation above may not be appropriate because tumor cells proliferate or die spontaneously and the simple Poisson model may not be appropriate to describe the kinetics. Hanin et al. (2001) proposed a new distribution for N that takes into account the spontaneous proliferation or death of tumor cells in fractionated radiotherapy. Here we will consider other distributions for N under the first-activation scheme (2.17) that lead to other cure models.

If N follows a Bernoulli distribution with the probability given in (2.2), then T defined in (2.17) will follow a mixture cure model with $S_u(t) = S^H(t)$. Thus, the mixture cure model is one of the models that can be defined using the first-activation scheme.

If N follows a binomial distribution with n and $e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}/(1+e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}})$ as the probability of success, then the survival function of T defined in (2.17) is (Peng and Xu, 2012)

$$\begin{aligned} S(t|\boldsymbol{z}) = P(N=0) + P(\tilde{T}_1 > t, \ldots, \tilde{T}_N > t, N \geq 1) &= \left(1+e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}\right)^{-n} \\ + \sum_{i=1}^{n} \left[S^H(t)\right]^i \binom{n}{i} \left(\frac{e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}}{1+e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}}\right)^i \left(1+e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}\right)^{i-n} \\ = \left(S^H(t)\frac{e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}}{1+e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}} + \frac{1}{1+e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}}\right)^n . \end{aligned} \tag{2.20}$$

This model is similar to the mixture cure model except it adds an extra parameter as the power to the unconditional survival function of the mixture cure model. This model arises when there are at most n carcinogenic cells in a patient. It is easy to see that the interpretation above reduces to the one for the mixture cure model when $n = 1$.

If N follows a geometric distribution with mean $\mu = e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}$, then the survival function of T defined in (2.17) is

$$\begin{aligned} S(t|\boldsymbol{z}) = P(N=0) + P(\tilde{T}_1 > t, \ldots, \tilde{T}_N > t, N \geq 1) &= \left(1+e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}\right)^{-1} \\ + \sum_{i=1}^{\infty} \left[S^H(t)\right]^i \left(\frac{e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}}{1+e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}}\right)^i \left(1+e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}\right)^{-1} \\ = \left(1+F^H(t)e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}\right)^{-1} . \end{aligned} \tag{2.21}$$

This is a PO model similar to (2.11) except that the baseline odds function is $F^H(t)$, which is an improper odds function. Thus, this model is an analogy

of the extension of the PH model to the PHC model by using an improper baseline odds function in the PO model. The resulting survival function $S(t|\boldsymbol{z})$ is an improper survival function and can be used to model survival data with a cured fraction. It is also called the proportional odds cure (POC) model. The POC model is considered in Tsodikov (2002); Gu et al. (2011), and covariates can also be included in $F^H(t)$ in the model.

It is clear that (2.17) always defines a cure model as long as N follows a distribution that has a probability mass at 0. We only considered discrete distributions above because of the biological interpretation attached to N. A non-discrete distribution can be considered as well, and this will be further discussed in the following section.

With the different cure models, it becomes important to choose an appropriate cure model among them for a given data set. The biological interpretation specified in (2.17) may help if it is plausible. However, the elusive nature of N and the latent variables $\tilde{T}_1, \ldots, \tilde{T}_N$ used in the biological interpretation makes this task challenging.

2.4.3 Cure Models Based on Frailty Models

A cure model can also be formed using the transformation model with a frailty (Choi et al., 2014). Given a frailty ξ, let the conditional cumulative hazard function be

$$H(t|\xi, \boldsymbol{x}) = \xi g\left(\int_0^t I(T \geq s)\exp(\boldsymbol{x}(s)'\boldsymbol{\gamma})dH_0(s)\right), \tag{2.22}$$

where $H_0(t)$ is a proper baseline cumulative hazard function and $g(\cdot)$ is a prespecified nonnegative transformation function that is strictly increasing and continuously differentiable and satisfies $g(0) = 0$, $g(\infty) = \infty$, and $g'(0) > 0$. This model will be suitable for modeling survival data with a cured fraction if ξ follows a distribution that has a probability mass at zero since the hazard for cured subjects is zero. For example, if ξ follows a Bernoulli distribution with probability π, then the marginal survival function from model (2.22) is

$$S(t|\boldsymbol{x}) = 1 - \pi + \pi \exp\left[-g\left(\int_0^t I(T \geq s)\exp(\boldsymbol{x}(s)'\boldsymbol{\gamma})dH_0(s)\right)\right].$$

This is a mixture model discussed in Section 2.2 if π is allowed to be affected by $\boldsymbol{z}$ in the logit form.

If ξ follows a Poisson distribution with mean μ, the marginal survival function from model (2.22) is

$$S(t|\boldsymbol{x}) = \exp\left[-\left(1 - e^{-g\left(\int_0^t I(T\geq s)\exp(\boldsymbol{x}(s)'\boldsymbol{\gamma})dH_0(s)\right)}\right)\mu\right].$$

This is the PHC model discussed in Section 2.4.1 if μ is allowed to be affected by $\boldsymbol{z}$ in the complementary log-log form and $\boldsymbol{\gamma} = 0$.

If ξ follows a geometric distribution with mean μ, the marginal survival function from model (2.22) is

$$S(t|\boldsymbol{x}) = 1 \Big/ \left[1 + \left(1 - e^{-g\left(\int_0^t I(T \geq s)\exp(\boldsymbol{x}(s)'\boldsymbol{\gamma})dH_0(s)\right)}\right)\mu\right].$$

This is essentially the POC model (2.21) if μ is allowed to be affected by $\boldsymbol{z}$ in a form of $\mu = \exp(\boldsymbol{z}'\gamma)$ and $\boldsymbol{\gamma} = 0$.

A more complicated distribution can be assumed for model (2.22) to be a cure model. In addition to discrete distributions considered above, a mixed type distribution with a continuous component and a discrete component at zero can be considered as well. One such a distribution is the compound Poisson distribution (Aalen, 1992). Such a distribution can also be formed by assuming ξ is a product of two random variables: one is a discrete with a positive probability mass at 0 and the other is a continuous distribution.

It is easy to see the cure models built on (2.22) are similar to those built on (2.17). This is not an unexpected result. Both the frailty ξ and N are latent variables and played a similar role in the models. They both cannot be observed. However, N in (2.17) provides an interpretation for ξ in (2.22).

2.4.4 Cure Models Based on Box-Cox Transformation

Another approach to build a general cure model is to use the Box-Cox transformation (Box and Cox, 1964). For example, the unconditional survival function of T may be defined as follows (Yin and Ibrahim, 2005a)

$$S(t|\boldsymbol{z},\boldsymbol{x}) = \begin{cases} \left\{1 - \dfrac{\lambda\exp(\gamma_0 + \boldsymbol{\gamma}'\boldsymbol{z})}{1 + \lambda\exp(\gamma_0 + \boldsymbol{\gamma}'\boldsymbol{z})}F^H(t|\boldsymbol{x})\right\}^{1/\lambda} & 0 < \lambda \leq 1 \\ \exp\left[-e^{\gamma_0 + \boldsymbol{\gamma}'\boldsymbol{z}}F^H(t|\boldsymbol{x})\right] & \lambda = 0, \end{cases} \tag{2.23}$$

where λ is the transformation parameter and $F^H(t|\boldsymbol{x})$ is a proper cumulative distribution function that may depend on a vector of covariates $\boldsymbol{x}$. It is easy to see that model (2.23) becomes the mixture cure model (2.1) when $\lambda = 1$ and the PHC model (2.18) when $\lambda \to 0$. Therefore model (2.23) unifies the mixture cure model and the PHC model.

This model can also be motivated using the model (2.20) (Peng and Xu, 2012). Assume temporarily that $1/\lambda$ is a positive integer and let N be a binomial variate with $n = 1/\lambda$ and $\lambda e^{\gamma_0 + \boldsymbol{\gamma}'z}/(1 + \lambda e^{\gamma_0 + \boldsymbol{\gamma}'z})$ as the probability of success. Then the survival function in (2.20) can be rewritten as

$$S(t|\boldsymbol{x},\boldsymbol{z}) = \left(1 - \frac{\lambda e^{\gamma_0 + \boldsymbol{\gamma}'\boldsymbol{z}}}{1 + \lambda e^{\gamma_0 + \boldsymbol{\gamma}'\boldsymbol{z}}}F^H(t|\boldsymbol{x})\right)^{1/\lambda},$$

which is the same as the survival function defined in (2.23). Thus, when $1/\lambda$ is a positive integer, this model arises when there are at most $1/\lambda$ carcinogenic

cells in a patient, and it generalizes the tumor kinetics interpretation above by allowing the number of carcinogenic cells $1/\lambda$ to be a real valued number in $[1, \infty)$. It is easy to see that the interpretation above reduces to the one for the mixture cure model when $\lambda = 1$.

Given the interpretation of $1/\lambda$ as the real-valued maximum number of carcinogenic cells that a patient can have, Peng and Xu (2012) also suggested to relax the requirement of $1/\lambda > 1$ by assuming $1/\lambda > 0$. That is,

$$S(t|\boldsymbol{z}, \boldsymbol{x}) = \begin{cases} \left\{1 - \dfrac{\lambda \exp(\gamma_0 + \boldsymbol{\gamma}'\boldsymbol{z})}{1 + \lambda \exp(\gamma_0 + \boldsymbol{\gamma}'\boldsymbol{z})} F^H(t|\boldsymbol{x})\right\}^{1/\lambda} & \lambda > 0 \\ \exp\left[-e^{\gamma_0 + \boldsymbol{\gamma}'\boldsymbol{z}} F^H(t|\boldsymbol{x})\right] & \lambda = 0. \end{cases} \tag{2.24}$$

Treating $1/\lambda$ as a positive real number is particularly attractive in understanding tumor kinetics. For example, the positive real number $1/\lambda$ may be considered as tumor volume in a patient, which may be more practical in cancer oncology than the number of tumor cells. An immediate result of extending λ from $(0, 1]$ to $(0, \infty)$ is that $S_u(t) > S^H(t|x)$ when $\lambda > 1$, a feature that Cooner et al. (2007) tried to obtain by defining the so-called last-activation scheme $T = \max\{\tilde{T}_1, \ldots, \tilde{T}_N\}$. The model (2.24) can be viewed as a model that unifies both the first- and last-activation schemes to allow $S_u(t|\boldsymbol{x})$ to be less than or greater than $S^H(t|\boldsymbol{x})$.

Another family of cure models is given by (Taylor and Liu, 2007)

$$S(t|\boldsymbol{z}, \boldsymbol{x}) = \begin{cases} \left\{1 + \left[\left(1 + e^{\gamma_0 + \boldsymbol{\gamma}'\boldsymbol{z}}\right)^{-\lambda} - 1\right] F^H(t|\boldsymbol{x})\right\}^{\frac{1}{\lambda}} & \lambda \neq 0 \\ (1 + e^{\gamma_0 + \boldsymbol{\gamma}'\boldsymbol{z}})^{-F^H(t|\boldsymbol{x})} & \lambda = 0. \end{cases} \tag{2.25}$$

It is easy to see that model (2.25) is a mixture cure model when $\lambda = 1$ and a PHC model when $\lambda = 0$ (except that the cure rate is $(1 + e^{\gamma_0 + \boldsymbol{\gamma}'\boldsymbol{z}})^{-1}$ instead of $\exp(-e^{\gamma_0 + \boldsymbol{\gamma}'\boldsymbol{z}})$). Note also that this model has the appealing feature that the limit as $t \to \infty$ of $S(t|\boldsymbol{z}, \boldsymbol{x})$ is $(1 + e^{\gamma_0 + \boldsymbol{\gamma}'\boldsymbol{z}})^{-1}$ and it does not depend on the value of λ.

Zeng et al. (2006) proposed another cure model based on the Box-Cox transformation as follows:

$$S(t|\boldsymbol{z}) = \begin{cases} [1 + \lambda e^{\gamma_0 + \boldsymbol{z}'\boldsymbol{\gamma}} F^H(t|\boldsymbol{x})]^{-1/\lambda} & \lambda > 0 \\ \exp[-e^{\gamma_0 + \boldsymbol{z}'\boldsymbol{\gamma}} F^H(t|\boldsymbol{x})] & \lambda = 0. \end{cases} \tag{2.26}$$

It is easy to show that model (2.26) unifies the PHC model (2.18) (when $\lambda \to 0$) and the POC model (2.21) (when $\lambda = 1$).

The Box-Cox transformation can also be used to unify the PHC model with a non-cure additive model as follows (Yin and Ibrahim, 2005b):

$$h(t|\boldsymbol{z}) = \begin{cases} [f^H(t)^\lambda + \lambda(\gamma_0 + \boldsymbol{z}'\boldsymbol{\gamma})]^{1/\lambda} & 0 < \lambda \leq 1 \\ f^H(t) \exp(\gamma_0 + \boldsymbol{z}'\boldsymbol{\gamma}) & \lambda = 0, \end{cases} \tag{2.27}$$

where $f^H(t)$ is a proper probability density function. It is clear that this model reduces to the PHC model when $\lambda = 0$ and an additive hazard model $h(t|\boldsymbol{z}) = f^H(t) + \gamma_0 + \boldsymbol{z}'\boldsymbol{\gamma}$ when $\lambda = 1$. The latter is the usual additive hazard model (Lin and Ying, 1994) except that the baseline hazard function is replaced with $f^H(t)$. When estimating this model, one has to ensure that the estimated hazard function $h(t|\boldsymbol{z})$ is non-negative.

There are a variety of estimation methods proposed in the literature to estimate the parameters in the unified cure models above, including parametric, semiparametric, and Bayesian methods. Unfortunately, due to the complexity of the unified models, the estimation methods usually fix the value of λ in the Box-Cox transformation at a prespecified value and then apply the estimation methods to estimate other parameters in the models. This approach greatly limits the application of the unified cure models.

2.5 Model Assessment

2.5.1 Choosing an Appropriate Parametric Distribution

One important issue with a parametric cure model is to determine a distribution for the cure model that is most appropriate for a data set. Preferably the subject knowledge can help determine the most appropriate distribution for a particular data set. However, if such knowledge is not available, statistical methods may be used to aid the selection among a few candidate distributions.

If the candidate distributions are nested, statistical tests may be used to select a parsimonious distribution that provides a fit to data with no significant difference from a fit by a complex distribution. For example, the generalized F distribution (2.9) reduces to the generalized log-gamma distribution (2.8) if $s_1 \to \infty$ or $s_2 \to \infty$, and the latter further reduces to the normal distribution if $q \to 0$ and the extreme value distribution if $q = 1$. Thus, it is possible to test adequacy of, say lognormal distribution, using the generalized F or generalized gamma distributions. Peng et al. (2001) discussed the use of the likelihood ratio test (LRT) within the generalized gamma distribution family to select a parsimonious distribution in the family for a data set. One challenging issue with this approach is that the asymptotic null distribution of the LRT statistic is no longer chi-square distribution if a parsimonious distribution corresponds to a boundary value of a parameter in the distribution family that it belongs to, such as the generalized log-gamma distribution versus generalized F distribution.

For non-nested distributions, some information criteria may be used to compare the cure models based on the distributions. The most common criteria include the Akaike information criterion (AIC), defined as

$$\text{AIC} = 2p - 2\ell(\hat{\boldsymbol{\theta}}),$$

and the Bayesian information criterion (BIC), defined as

$$\text{BIC} = p\log(n) - 2\ell(\hat{\boldsymbol{\theta}}),$$

where p is the number of estimated parameters in the cure model. A cure model with a smaller value of AIC or BIC is preferred to a model with a larger value. This method, however, may not be useful if one is interested in the adequacy of one model relative the other model for data.

2.5.2 Mixture vs Non-Mixture Cure Models

When $\boldsymbol{z}$ is absent and $F^H(t|\boldsymbol{x})$ is unspecified, the mixture cure and PHC models are equivalent. They are simply different forms of the same model. When $\boldsymbol{z}$ is present and $\boldsymbol{x}$ is absent, the PHC model is a PH model and the mixture cure model is not, and they are clearly models for different data structures. When both $\boldsymbol{x}$ and $\boldsymbol{z}$ are present, both models are flexible and they can be considered for modeling survival data with a cure fraction. An emerging question is to determine which one of the two models is appropriate for given data, particularly when there is no convincing biological evidence to favor any of the two models. Similar questions apply to other cure models. With the unified cure models discussed in Section 2.4.4, it is possible to use statistical tests to examine the adequacy of models relative to the unified cure models and choose one that does not show a significant difference (Peng and Xu, 2012). For example, if we consider the following hypotheses under the model (2.24):

$$H_0 : \lambda = 1 \text{ vs } H_a : \lambda \neq 1, \tag{2.28}$$

$$H_0 : \lambda = 0 \text{ vs } H_a : \lambda > 0, \tag{2.29}$$

then the first set is to test the adequacy of the mixture cure model and the second set is to test the adequacy of the PHC model. Let $\boldsymbol{\theta} = (\lambda, \boldsymbol{\gamma}', \boldsymbol{\beta}', \boldsymbol{\alpha}')'$ be the parameters in model (2.24) ($\boldsymbol{\alpha}$ is a set of parameters in the baseline distribution in $F^H(t|\boldsymbol{x})$), $\ell(\lambda, \boldsymbol{\gamma}, \boldsymbol{\beta}, \boldsymbol{\alpha})$ be the log-likelihood function from the model, $\hat{\boldsymbol{\theta}} = \arg\max_{(\lambda,\boldsymbol{\gamma},\boldsymbol{\beta},\boldsymbol{\alpha})} \ell(\lambda, \boldsymbol{\gamma}, \boldsymbol{\beta}, \boldsymbol{\alpha})$, $\hat{\boldsymbol{\theta}}_0 = \arg\max_{(\lambda=0,\boldsymbol{\gamma},\boldsymbol{\beta},\boldsymbol{\alpha})} \ell(\lambda, \boldsymbol{\gamma}, \boldsymbol{\beta}, \boldsymbol{\alpha})$, and $\hat{\boldsymbol{\theta}}_1 = \arg\max_{(\lambda=1,\boldsymbol{\gamma},\boldsymbol{\beta},\boldsymbol{\alpha})} \ell(\lambda, \boldsymbol{\gamma}, \boldsymbol{\beta}, \boldsymbol{\alpha})$. The LRTs for the two sets of hypotheses are

$$X^2_{MCM} = -2[\ell(\hat{\boldsymbol{\theta}}_1) - \ell(\hat{\boldsymbol{\theta}})], \quad X^2_{PHC} = -2[\ell(\hat{\boldsymbol{\theta}}_0) - \ell(\hat{\boldsymbol{\theta}})] \tag{2.30}$$

and the score tests are

$$\begin{aligned} Z^2_{MCM} &= (\partial\ell(\boldsymbol{\theta})/\partial\lambda)^2 \Big/ \left(-\partial^2\ell(\boldsymbol{\theta})/\partial\lambda^2 - AB^{-1}A^T\right)\Big|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}_1}, \\ Z^2_{PHC} &= (\partial\ell(\boldsymbol{\theta})/\partial\lambda)^2 \Big/ \left(-\partial^2\ell(\boldsymbol{\theta})/\partial\lambda^2 - AB^{-1}A^T\right)\Big|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}_0}, \end{aligned} \tag{2.31}$$

where $A = \left(-\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial\lambda\partial\boldsymbol{\gamma}^T}, -\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial\lambda\partial\boldsymbol{\beta}^T}, -\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial\lambda\partial\boldsymbol{\alpha}^T}\right)$ and

$$B = \begin{pmatrix} -\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial\boldsymbol{\gamma}\partial\boldsymbol{\gamma}^T} & -\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial\boldsymbol{\gamma}\partial\boldsymbol{\beta}^T} & -\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial\boldsymbol{\gamma}\partial\boldsymbol{\alpha}^T} \\ -\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\gamma}^T} & -\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^T} & -\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial\boldsymbol{\beta}\partial\boldsymbol{\alpha}^T} \\ -\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial\boldsymbol{\alpha}\partial\boldsymbol{\gamma}^T} & -\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial\boldsymbol{\alpha}\partial\boldsymbol{\beta}^T} & -\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial\boldsymbol{\alpha}\partial\boldsymbol{\alpha}^T} \end{pmatrix}.$$

For H_0 in (2.28), $\lambda = 1$ is an interior point of the parameter space of λ in (2.24). Therefore, based on the standard likelihood theory both the LRTs and the score tests approximately follow the chi-square distribution with 1 degree of freedom when the H_0 is true. For H_0 in (2.29), $\lambda = 0$ is on the boundary of the parameter space of λ. The null distribution of the score test can still be approximated by the chi-square distribution with 1 degree of freedom (Rao, 2005). The null distribution of the LRT under H_0 is however nonstandard (Self and Liang, 1987). To overcome this issue, Peng and Xu (2012) suggested to extend the domain of λ in (2.24) further into negative values by $\lambda > -\exp(-\boldsymbol{\gamma}\boldsymbol{z})$ so that $\lambda = 0$ is not on the boundary and the standard LRT results can apply.

The tests have reasonable power to select between the mixture cure model and the PHC model if $F^H(t)$ is parametrically specified. It is expected that other unified cure models have similar properties.

2.5.3 Goodness of Fit by Residuals

There are not many methods proposed specifically for checking the goodness of fit of parametric cure models. One method is to use a revised Schoenfeld residual (Wileyto et al., 2013) to compare different parametric mixture cure models (2.1). Assuming the same set of covariates in both the incidence and latency parts of the mixture cure model, the revised Schoenfeld residual is defined in the following form

$$\delta_i \left[\boldsymbol{x}_i - \frac{\sum_{j=1}^n Y_j(t_i)\boldsymbol{x}_j h(t_i|\boldsymbol{x}_j, \boldsymbol{z}_j)}{\sum_{j=1}^n Y_j(t_i) h(t_i|\boldsymbol{x}_j, \boldsymbol{z}_j)} \right], \quad i = 1, \ldots, n,$$

where $h(t|\boldsymbol{x}, \boldsymbol{z})$ is the unconditional hazard function corresponding to $S(t|\boldsymbol{x}, \boldsymbol{z})$ in (2.1), $Y_i(t) = I(t_i \geq t)$, and $I(A)$ is the indicator function: $I(A) = 1$ if A is true and $I(A) = 0$ otherwise. This residual measures the difference between the covariate values of a subject with uncensored times and the weighted average of the covariate values of all subjects who are still at the risk at the uncensored time, and the unconditional hazard function at the uncensored time is used as the weight. The residuals from a correctly specified mixture cure model have zero mean and they are useful to examine the overall fit of a mixture cure model (2.1). A plot of the residuals against time is useful to reveal any patterns of the fit. The patterns can also be revealed by fitting a polynomial regression to the residuals against time.

Section 3.5 will introduce methods for assessing the fit of semiparametric and nonparametric cure models. Some of them may be used for parametric cure models too.

2.6 Software and Applications

For some simple parametric cure models, special software packages may not be required and standard statistical packages often can be used to fit them with minimal coding work. For example, a simple mixture cure model without covariates such as (2.1) with $S_u(t)$ from the exponential or Weibull distribution can be easily coded in R (Ihaka and Gentleman, 1996) or SAS (SAS Institute Inc., 2011) to obtain the maximum likelihood estimates of parameters in the model. When a complicated distribution is considered for $S_u(t)$ or nonlinear covariate effects are considered in π or in $S_u(t)$, the necessary coding becomes complicated in standard statistical software packages and specialized software packages are useful to fit such models. We will show some R software packages and SAS macros for parametric cure models discussed in this chapter and illustrate their applications to real data sets for some cure models.

The real data we consider are from a comparative study of autologous and allogeneic transplantation in patients with high-risk acute lymphoblastic leukemia from March 1982 to May 1987 (Kersey et al., 1987). In this study, 46 patients who had sibling donors HLA-matched by mixed-lymphocyte culture received allogeneic marrow transplants, and 45 Patients without a matched sibling donor and with the common B-lineage phenotypes received autologous marrow purged with a mixture of the three antibodies BA-I, BA-2, and BA-3 plus complement. One objective of the study is to assess the time to leukemia recurrence in recipients of autologous and allogeneic transplants. Some patients did not experience a relapse due to either death before a relapse or insufficient follow-up before a relapse, and their observed times are censored at the time of death or at the time of last observation. There are total 22 censored times, 13 from patients with allogeneic transplants and 9 from those with autologous transplants. The data are displayed in Table 2.1. The numbers followed by a plus sign "+" are censored times.

To demonstrate the potential presence of cured subjects in this study, we estimate the survival functions of the two groups of data with the Kaplan-Meier survival estimator (Kaplan and Meier, 1958) and plot the survival curves in Figure 2.4. The small circles on the curves mark the censored times. It can be seen that most of relapses occurred within 500 days and there is no more relapse after 1256 days until 1845 days, whereas the censoring occurred only after 500 days. The obvious plateaus in their right tails and the long follow-up in the study strongly suggest the presence of cured subjects among the study subjects, and a cure model should be considered for the data. The problems of

TABLE 2.1
Time to leukemia recurrence or relapse (days) for leukemia patients with marrow transplants

Allogeneic	Autologous
11, 14, 23, 31, 32, 35, 51, 59, 62, 78, 78, 79, 87, 99, 100, 141, 160, 166, 216, 219, 235, 250, 270, 313, 332, 352, 368, 468, 491, 511, 557, 628+, 726+, 819, 915+, 966+, 1109+, 1158+, 1256, 1614+, 1619+, 1674+, 1712+, 1745+, 1820+, 1825+	21, 40, 42, 50, 53, 54, 56, 61, 64, 67, 73, 76, 79, 81, 88, 95, 98, 98, 99, 104, 105, 106, 112, 131, 147, 171, 172, 179, 189, 195, 199, 213, 223, 224, 277, 724+, 729+, 734, 1053+, 1094+, 1192+, 1475+, 1535+, 1535+, 1845+

interest for this data include estimating cure rates in the two groups and assessing the significance of the differences between the two marrow transplants in cure rates and in survival times of the uncured patients.

2.6.1 R Package gfcure

The R package `gfcure` based on the work of Peng et al. (1998) can fit various parametric AFT mixture cure models. We use `gfcure` function in the package to fit lognormal AFT mixture cure model to the leukemia data with `transplant = 1` for Autologous group and `transplant = 0` for Allogeneic group as follows:

```
> summary(gfcure(Surv(time, cens) ~ transplant,  ~ transplant, data
+ = goldman.data, dist = "lognormal"))

Lognormal mixture model

The maximum loglikelihood is -142.4011

Terms in the accelerated failure time model:
                     Coefficients   Std.err   z-score   p-value
Log(scale)              -0.034372 0.0949418 -0.362032 0.7173281
(Intercept)              4.948055 0.1758938 28.130923 0.0000000
transplantAutologous    -0.280250 0.2406373 -1.164618 0.2441738

Terms in the logistic model:
                     Coefficients   Std.err   z-score   p-value
(Intercept)              0.990842 0.3434634  2.884853 0.0039160
transplantAutologous     0.439686 0.5162291  0.851726 0.3943664
```

The syntax of `gfcure` is similar to `survreg` in R with two formula arguments, one for the latency submodel and one for the incidence submodel, and the

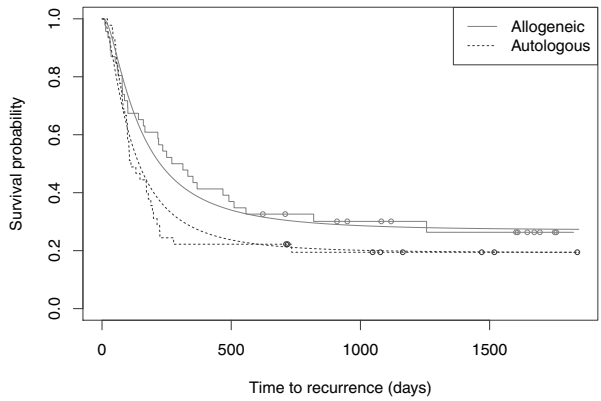

FIGURE 2.4
Estimated survival curves from the lognormal AFT mixture cure model (smooth curves) and the Kaplan-Meier method (step lines) for the two groups of leukemia patients with marrow transplants.

latter does not specify any variable to the left of ~. The estimated $\hat{\boldsymbol{\gamma}} = (0.99, 0.44)'$ and $\hat{\boldsymbol{\beta}} = (4.95, -0.28)'$. The estimates imply that the estimated cure rate is $1/(1 + e^{0.99}) = 0.27$ for Allogeneic group and $1/(1 + e^{0.99+0.44}) = 0.19$ for Autologous group. However, the difference in cure rate is not statistically significant. The estimated coefficient of `transplant` in the latency submodel is -0.28, which indicates that for an uncured patient, the mean log survival time in Autologous group is about 0.28 shorter than the mean log survival time in Allogeneic group, or the mean survival time of Autologous group is $e^{-0.28} = 0.76$ times of mean survival time of Allogeneic group. Again the difference in the latency submodel is not statistically significant. To examine the fit, the estimated survival curves can be obtained are plotted using `predict.gfcure` and `plot.predict.gfcure` functions as follows:

```
> z = gfcure(Surv(time, cens) ~ transplant,  ~ transplant, data =
+ goldman.data, dist = "lognormal")
> plot(survfit(Surv(time, cens) ~ transplant, data = goldman.data),
+ lty = 1:2, col = c(2,4), mark = 21, ylab = "Survival
+ probability", xlab = "Time to recurrence (days)")
> lines(predict(z, newdata = goldman.data[c(1, 91), ], newtime =
+ (1:1000)*1.845), showcure = T, col = c(2, 4))
```

The curves are plotted along with the Kaplan-Meier curves in Figure 2.4. It is easy to see that the survival curves from the lognormal AFT mixture cure models agree with the survival curves from the Kaplan-Meier survival estimator at most of time except the period around 200 to 500 days where the cure model tends to underestimate the survival probability for Allogeneic group and overestimate the survival probability for Autologous group.

Other distributions can be considered in `gfcure` for the latency part. They include `exponential`, `rayleigh`, `weibull`, `gamma`, `loglogistic`, `gll` (generalized log-logistic), `gf` (generalized F), and `egg` (extended generalized gamma). Following are `gfcure` with `exponential`, `weibull`, `gamma`, and `egg` for the leukemia data:

```
> gfcure(Surv(time, cens) ~ transplant, ~ transplant, data =
+ goldman.data, dist = "exponential")

Call:
gfcure(formula = Surv(time, cens) ~ transplant, cureform =
    ~transplant, data = goldman.data, dist = "exponential")

Distribution: Exponential

Coefficients:
         (Intercept) transplantAutologous          (Intercept)
           5.5399865           -0.6462144            0.9892163
transplantAutologous
           0.4022781

Maximized Loglikelihood: -146.0479

> gfcure(Surv(time, cens) ~ transplant, ~ transplant, data =
+ goldman.data, dist = "weibull")

Call:
gfcure(formula = Surv(time, cens) ~ transplant, cureform =
    ~transplant, data = goldman.data, dist = "weibull")

Distribution: Weibull

Coefficients:
          Log(scale)          (Intercept) transplantAutologous
          -0.1346419            5.5849274           -0.6559098
         (Intercept) transplantAutologous
           0.9708204            0.4169098

Maximized Loglikelihood: -145.0319

> gfcure(Surv(time, cens) ~ transplant, ~ transplant, data =
+ goldman.data, dist = "gamma")

Call:
gfcure(formula = Surv(time, cens) ~ transplant, cureform =
    ~transplant, data = goldman.data, dist = "gamma")

Distribution: Gamma
```

TABLE 2.2
Parametric mixture cure models with different latency distributions for the leukemia data

Latency distribution	Maximum log-likelihood	AIC	BIC	LRT	df	P-value
Generalized gamma	−142.3	296.6	311.7	-	-	-
Lognormal	−142.4	294.8	307.4	0.154	1	0.695
Weibull	−145.0	300.1	312.6	5.415	1	0.020
Gamma	−144.2	298.4	310.9	3.729	1	0.053
Exponential	−146.0	300.1	310.1	7.447	2	0.024

```
Coefficients:
               Shape          (Intercept) transplantAutologous
           0.3056717            5.2088573           -0.6249223
         (Intercept) transplantAutologous
           0.9662783            0.4216691

Maximized Loglikelihood: -144.1888

> gfcure(Surv(time, cens) ~ transplant, ~ transplant, data =
+ goldman.data, dist = "egg")

Call:
gfcure(formula = Surv(time, cens) ~ transplant, cureform =
    ~transplant, data = goldman.data, dist = "egg")

Distribution: Extended Generalized Gamma

Coefficients:
               Shape           Log(scale)          (Intercept)
          0.12433492          -0.04766131           5.03219942
transplantAutologous          (Intercept) transplantAutologous
         -0.34089158           0.98299111           0.43270369

Maximized Loglikelihood: -142.3244
```

Without a priori knowledge about the distribution, one needs to determine which parametric distribution is the most suitable as the latency distribution. One can use the methods discussed in Section 2.5 to select a distribution that provides an adequate fit to the data. Table 2.2 shows the fit of AFT mixture cure models above along with AIC and BIC values. Since the first distribution includes the rest as special cases, the LRTs to test the adequacy of the special cases relative to the first distribution are also presented in the table. It is clear that the lognormal distribution is recommended over the Weibull, gamma and the exponential distributions when used as the baseline distribution in the AFT latency submodel for the data.

2.6.2 R Package mixcure

The R package `mixcure` is designed to fit the mixture cure model by using existing R packages to fit the latency and incidence submodels of the cure model. The parametric mixture cure model can be fit by `mixcure` by using `survreg` in R package `survival` or `flexsurvreg` in R package `flexsurv` to fit the latency submodel and `glm` to fit the incidence submodel. We first use `survreg` in `mixcure` to fit the latency submodel and `glm` to fit the incidence submodel of the mixture cure model for the leukemia data. The latency distribution can be any of the distributions that are available in `survreg`. We only consider the lognormal distribution as an example.

```
> summary(mixcure(Surv(time, cens) ~ transplant, ~ transplant, data
+ = goldman.data, lmodel = list(fun = "survreg", dist =
+ "lognormal"), savedata = T))

Call:
mixcure(lformula = Surv(time, cens) ~ transplant, iformula =
    ~transplant, data = goldman.data, lmodel =
        list(fun = "survreg", dist = "lognormal"), savedata = T)

Latency model for uncured:
                     coefficient    stderr    zscore    pvalue
(Intercept)            4.9480436 0.2168121 22.821806 0.0000000
transplantAutologous  -0.2802129 0.2322965 -1.206273 0.2277124

Incidence model:
                     coefficient    stderr    zscore     pvalue
(Intercept)            0.9908102 0.3901210 2.5397510 0.01109314
transplantAutologous   0.4398913 0.5904678 0.7449878 0.45627909
```

The standard errors of the estimates are obtained by a bootstrap method when using `summary` to report the results. The results are similar to the results from `gfcure`. Note that the default model for the incidence part is `glm` with binomial distribution and `logit` link function and thus it is not specified in the above command. If one is interested in fitting a complementary log-log (2.3) instead of a logistic model (2.2) for the incidence part and the latency part stays unchanged, the model can be fitted as follows:

```
> mixcure(Surv(time, cens) ~ transplant, ~ transplant, data =
+ goldman.data, lmodel = list(fun = "survreg", dist = "lognormal"),
+ imodel = list(fun = "glm", family = binomial(link = "cloglog")),
+ savedata = T)

Call:
mixcure(lformula = Surv(time, cens) ~ transplant, iformula =
    ~transplant, data = goldman.data, lmodel =
                list(fun = "survreg", dist = "lognormal"),
    imodel = list(fun = "glm", family = binomial(link = "cloglog")),
    savedata = T)
```

```
Latency model for uncured:
                     coefficient    stderr    zscore    pvalue
(Intercept)            4.9480436 0.2270397 21.793734 0.0000000
transplantAutologous  -0.2802129 0.2659556 -1.053608 0.2920626

Incidence model:
                     coefficient    stderr    zscore    pvalue
(Intercept)            0.2673914 0.1707122 1.5663289 0.1172717
transplantAutologous   0.2304219 0.2699945 0.8534319 0.3934198
```

Other link functions, such as probit link, are also available.

`flexsurvreg` can also be used to fit parametric latency submodels with various parametric distributions in `mixcure`, including custom distributions by users. Particularly, `flexsurvspline`, a wrapper around `flexsurvreg`, can fit parametric PH, PO, and probit latency submodels with a spline-based baseline distribution by dynamically constructing a custom spline-based survival distribution. We consider to use `mixcure` to fit the PH mixture cure model (2.5) and the PO mixture cure model (2.11) for the leukemia data by using `flexsurvreg` with the log cumulative baseline hazard function and the log cumulative baseline odds function modeled by splines with 1 knot:

```
> # PH mixture cure model
> zph = mixcure(Surv(time, cens) ~ transplant, ~ transplant, data =
+ goldman.data, lmodel = list(fun = "flexsurvspline", k=1,
+ scale="hazard"))
> zph

Call:
mixcure(lformula = Surv(time, cens) ~ transplant, iformula =
    ~transplant,  data = goldman.data, lmodel = list(fun =
"flexsurvspline", k = 1, scale = "hazard"), savedata = T, debug = T)

Latency model:
Call:
flexsurvspline(formula = as.formula(ff), data = data[data$postuncure>
    0, ], weights = postuncure, k = 1, scale = "hazard")

Estimates:
                    data mean  est       L95%      U95%      se
gamma0                   NA   -10.4170  -13.4723   -7.3618    1.5588
gamma1                   NA     2.1669    1.4596    2.8742    0.3609
gamma2                   NA     0.0954    0.0403    0.1504    0.0281
transplantAutologous 0.4945     0.6656    0.1631    1.1681    0.2564
                    exp(est)  L95%      U95%
gamma0                     NA        NA        NA
gamma1                     NA        NA        NA
gamma2                     NA        NA        NA
transplantAutologous    1.9457    1.1771    3.2160
```

```
N = 91,  Events: 69,  Censored: 22
Total time at risk: 41432
Log-likelihood = -425.8415, df = 4
AIC = 859.6829

Incidence model:

Call:  structure(expression(glm(formula = postuncure ~ transplant,
data = data, ...)), srcfile = <environment>, wholeSrcref =
structure(c(1L, 0L, 2L, 0L, 0L, 0L, 1L, 2L),
srcfile = <environment>, class = "srcref"))

Coefficients:
         (Intercept)  transplantAutologous
              1.0899                0.3131

Degrees of Freedom: 90 Total (i.e. Null);  89 Residual
Null Deviance:      87.37
Residual Deviance: 86.98        AIC: 104

EM algorithm finished  59 iterations with final error 9.344802e-05

> # PO mixture cure model
> zpo = mixcure(Surv(time, cens) ~ transplant, ~ transplant, data =
+ goldman.data, lmodel = list(fun = "flexsurvspline", k=1,
+ scale="odds"))
> zpo

Call:
mixcure(lformula = Surv(time, cens) ~ transplant, iformula =
    ~transplant, data = goldman.data,
    lmodel = list(fun = "flexsurvspline", k = 1, scale = "odds"),
    savedata = T, debug = T)

Latency model:
Call:
flexsurvspline(formula = as.formula(ff),
    data = data[data$postuncure > 0, ],
    weights = postuncure, k = 1, scale = "odds")

Estimates:
                   data mean  est       L95%      U95%      se
gamma0                   NA   -11.9844  -15.4662   -8.5027    1.7764
gamma1                   NA     2.5649    1.7533    3.3765    0.4141
gamma2                   NA     0.1182    0.0541    0.1824    0.0327
transplantAutologous 0.4945     0.6525   -0.1205    1.4255    0.3944
                   exp(est)  L95%      U95%
gamma0                   NA        NA        NA
```

```
gamma1                   NA        NA        NA
gamma2                   NA        NA        NA
transplantAutologous 1.9203    0.8865    4.1599

N = 91,  Events: 69,  Censored: 22
Total time at risk: 41432
Log-likelihood = -453.9555, df = 4
AIC = 915.9111

Incidence model:

Call:  structure(expression(glm(formula = postuncure ~ transplant,
   data = data, ...)), srcfile = <environment>,
   wholeSrcref = structure(c(1L, 0L, 2L, 0L, 0L, 0L, 1L, 2L),
         srcfile = <environment>, class = "srcref"))

Coefficients:
         (Intercept)  transplantAutologous
             1.99655               0.08252

Degrees of Freedom: 90 Total (i.e. Null);  89 Residual
Null Deviance:      35.17
Residual Deviance: 35.15        AIC: 55.43

EM algorithm finished  6 iterations with final error 5.917918e-05
```

Note the standard errors and confidence intervals reported above are directly from `flexsurvspline` and `glm` and they do not provide correct standard errors and confidence intervals for the estimates in the mixture cure models. Correct standard errors and confidence intervals should be obtained using `summary` as we did above. The fit of the two models to the data can be examined by estimating the unconditional survival functions $S(t)$ from the models and plot them as follows:

```
> plot(survfit(Surv(time, cens) ~ transplant, data = goldman.data),
+ mark = 21, ylab = "Survival probability", xlab = "Time to
+ recurrence (days)")
> plot(predict(zph, newdata = goldman.data[c(1, 91), ], times =
+ 1:2000), add = T, lty = c(2,2))
> plot(predict(zpo, newdata = goldman.data[c(1, 91), ], times =
+ 1:2000), add = T, lty = c(3,3))
> legend("topright", legend = c("KM", "PH mixture cure", "PO
+ mixture cure"), lty = 1:3)
```

Figure 2.5 shows the estimated survival functions. It is clear that the PH mixture cure model fits the data better than the PO mixture cure model.

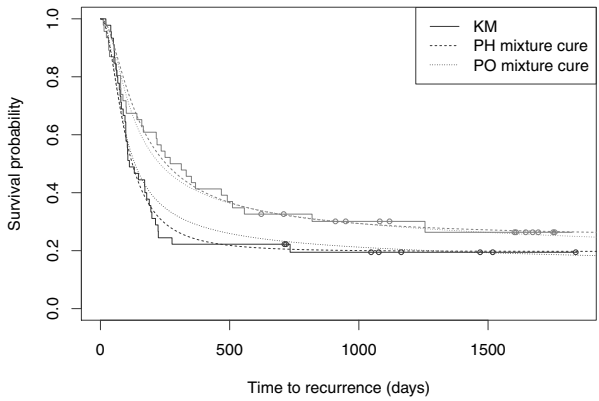

FIGURE 2.5
Estimated survival curves from the PH and PO mixture cure models with spline-based baseline distributions (smooth curves) and the Kaplan-Meier method (step lines) for the two groups (top: Allogeneic, bottom: Autologous) of leukemia patients with marrow transplants.

In addition to `survival` and `flexsurv`, it is possible to consider other R packages that can fit parametric survival models in `mixcure`. Check the manual of `mixcure` for details.

2.6.3 R Package flexsurvcure

The R package `flexsurvcure` is a wrapper for R package `flexsurv` to fit various parametric cure models, including both mixture and non-mixture cure models. It achieves this by providing custom distributions that correspond to cure models and supplying them to `flexsurv`. The feature of `flexsurv` to allow arbitrary-dimension distributions, such as spline-based hazard functions, can also be used in cure models when using `flexsurvcure`. The other feature of `flexsurv` to allow other parameters in a distribution to depend on covariates makes the resulting cure models more flexible than the parametric cure models discussed in previous sections. We first demonstrate how `flexsurvcure` can fit the lognormal mixture cure model to the leukemia data as follows:

```
> zm = flexsurvcure(Surv(time,cens) ~ transplant +
+ meanlog(transplant), data = goldman.data, dist = "lnorm", link =
+ "logistic")
> zm

Call:
flexsurvcure(formula = Surv(time, cens) ~ transplant +
    meanlog(transplant), data = goldman.data,
                 dist = "lnorm", link = "logistic")
```

```
Estimates:
                              data mean  est     L95%   U95%   se
theta                              NA    0.2708  0.1592 0.4212     NA
meanlog                            NA    4.9480  4.6033 5.2928 0.1759
sdlog                              NA    0.9662  0.8022 1.1638 0.0917
transplantAutologous           0.4945   -0.4399 -1.4517 0.5719 0.5162
meanlog(transplantAutologous) 0.4945   -0.2802 -0.7518 0.1914 0.2406
                              exp(est)  L95%     U95%
theta                              NA       NA       NA
meanlog                            NA       NA       NA
sdlog                              NA       NA       NA
transplantAutologous           0.6441   0.2342   1.7717
meanlog(transplantAutologous) 0.7556   0.4715   1.2110

N = 91,  Events: 69,  Censored: 22
Total time at risk: 41432
Log-likelihood = -471.7213, df = 5
AIC = 953.4426
```

The results are similar to the results from `gfcure`. However, the reported maximum likelihood, AIC and BIC values are different due to different parameterization.

We then consider to fit the data with the PHC model (2.18), a non-mixture cure model. To make the model more comparable with the mixture cure model above, we assume that $F^H(t)$ is from the lognormal distribution and the location parameter in the lognormal distribution is a linear function of the treatment group variable.

```
> znm = flexsurvcure(Surv(time, cens) ~ transplant +
+ meanlog(transplant), data = goldman.data, dist = "lnorm", link =
+ "loglog", mixture = FALSE)
> znm

Call:
flexsurvcure(formula = Surv(time, cens) ~ transplant +
    meanlog(transplant), data = goldman.data,
                dist = "lnorm", link = "loglog", mixture = FALSE)

Estimates:
                              data mean est     L95%   U95%   se
theta                              NA   0.2840  0.4211 0.1601     NA
meanlog                            NA   5.2861  4.8583 5.7140 0.2183
sdlog                              NA   1.0285  0.8350 1.2669 0.1094
transplantAutologous           0.4945   0.3712 -0.1631 0.9056 0.2726
meanlog(transplantAutologous) 0.4945  -0.0425 -0.5786 0.4936 0.2735
                              exp(est)  L95%    U95%
theta                              NA       NA      NA
meanlog                            NA       NA      NA
sdlog                              NA       NA      NA
```

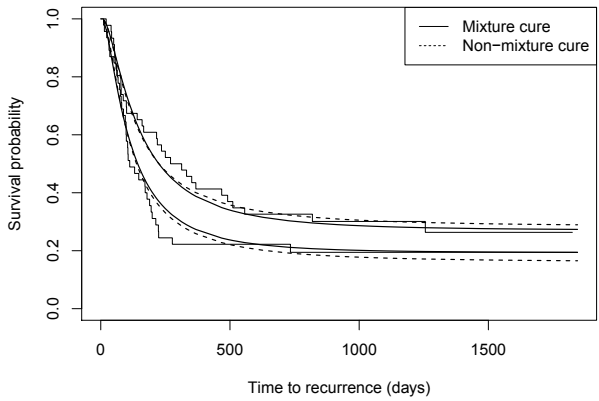

FIGURE 2.6
Estimated survival curves from the mixture cure and non-mixture cure models with lognormal distributions (smooth curves) and the Kaplan-Meier method (step lines) for the two groups (top: Allogeneic, bottom: Autologous) of leukemia patients with marrow transplants.

```
transplantAutologous          1.4495   0.8495  2.4733
meanlog(transplantAutologous) 0.9584   0.5607  1.6383

N = 91,  Events: 69,  Censored: 22
Total time at risk: 41432
Log-likelihood = -471.2055, df = 5
AIC = 952.411
```

The fit of the two models can be compared by plotting the estimated survival functions from the two models as follows:

```
> plot(zm, col = 1, ylab = "Survival probability", xlab = "Time to
+ recurrence (days)")
> plot(znm, add = T, lty = 2, col = 1)
> legend("topright", legend = c("Mixture cure", "Non-mixture
+ cure"), lty = 1:2)
```

The fitted survival curves are shown in Figure 2.6. It appears that the mixture model shows a slightly better fit to the right tail of the Kaplan-Meier survival curves than the non-mixture cure model.

2.6.4 SAS Macro PSPMCM

The SAS macro `PSPMCM` (Corbiere and Joly, 2007) implements the parametric AFT mixture cure model with the several distributions as the latency distribution, including `EXP` (exponential), `WEIBULL`, `LLOGISTIC`, and `LOGNORMAL`. We use the macro to fit the lognormal mixture cure model to the leukemia data:

```
%pspmcm(DATA=goldmandata,ID=id,CENSCOD=cens,TIME=time,
  VAR= transplant(IS), INCPART=logit, SURVPART=LOGNORMAL,
  TAIL=zero, SU0MET=pl, FAST=Y,BOOTSTRAP=Y,
  NSAMPLE=2000, STRATA=, MAXITER=200,CONVCRIT=1e-5,
  ALPHA=0.05, BASELINE=Y, BOOTMET=ALL, JACKDATA=,
  GESTIMATE=Y, SPLOT=Y, PLOTFIT=Y);
run;
```

```
                                  Fit Statistics

                     -2 Log Likelihood                  943.4
                     AIC (smaller is better)            953.4
                     AICC (smaller is better)           954.1
                     BIC (smaller is better)            966.0

                          results for goldmandata
                       distribution LOGNORMAL, link=LOGIT

                              Parameter Estimates

                          Standard                                 95% Confidence
  Parameter    Estimate     Error   DF  t Value   Pr > |t|        Limits      Gradient
  L_int          0.9908    0.3435   91     2.88     0.0049   0.3086   1.6730  -0.00003
  L_TRANSPLANT   0.4399    0.5162   91     0.85     0.3964  -0.5856   1.4654  -8.49E-6
  _scale         4.9480    0.1759   91    28.13     <.0001   4.5986   5.2974  -0.00002
  S_TRANSPLANT   0.2900    0.2499   91     1.16     0.2489  -0.2064   0.7864     -8E-6
  _shape         0.9662   0.09174   91    10.53     <.0001   0.7840   1.1485  0.000029
```

The results are similar to the results from `gfcure`. Again, the reported maximum likelihood, AIC and BIC values are different due to different parameterization.

2.7 Summary

In this chapter, we presented the basics of parametric mixture cure models and their estimation methods. Because of the finite number of parameters in the models, the maximum likelihood estimation method can be readily used to estimate the parameters in the models. In particular, we showed in detail how the EM algorithm is used to obtain the maximum likelihood estimates of the parameters in the parametric models. The EM algorithm is an important estimation method for mixture models and is widely used for cure models. The chapter also described some parametric non-mixture cure models and the cure models that unify the mixture cure models and non-mixture cure models. Some methods for model selection and goodness-of-fit assessment were also introduced. Finally, we demonstrated 3 R packages and 1 SAS macro that can be used to fit some parametric cure models. This chapter provides the foundation of cure models for the remaining chapters.

The mixture cure model is also known as a mover-stayer model in some fields. See Yamaguchi (2003) and Shen and Cook (2014) for some further discussions under the mover-stayer model.

Even though the application of parametric cure models may be limited by their strong parametric assumptions, they are still widely used because of their simple estimation methods and smoothed estimates of the latency survival functions that allow extrapolation.

We did not include any discussions on the asymptotic properties in this chapter because the asymptotic properties of the maximum likelihood estimates usually follow from the large sample likelihood theory.

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